

UML Use-Case Modelling & Structure

Airport Lost Luggage Claim & Tracking Portal (Problem Statement #30)

PES University - Dept. of CSE | SE Lab 1

Target System: Airport Lost Luggage Claim & Tracking Portal

Primary Actors: Passenger, Baggage Service Agent

1. System Actors

- **Passenger (Primary Actor):** End-user filing baggage claims and checking tracking timelines.
- **Baggage Service Agent (Secondary Actor):** Airport staff updating scan logs and processing exceptions manually.

2. Use Cases Summary

- **UC-01: Submit Lost Luggage Claim** (Passenger)
- **UC-02: Track Baggage Timeline** (Passenger)
- **UC-03: Verify Passenger Identity** (System Security)
- **UC-04: File Compensation Request** (Passenger)
- **UC-05: Update Baggage Location & Status** (Agent)
- **UC-06: Cross-Reference Unrouted Logs** (Agent)

3. Use-Case Diagram Relationships

- **«include» Relationship:**
UC-01 (Submit Claim) and UC-02 (Track Timeline) **«include»** UC-03 (Verify Passenger Identity).
Rationale: Every claim submission or tracking lookup strictly requires verifying passenger credentials prior to processing.
- **«extend» Relationship:**
UC-04 (File Compensation Request) **«extend»** UC-01 (Submit Claim).
Rationale: Compensation filing is an optional extension triggered only when baggage delay exceeds the 24-hour threshold.